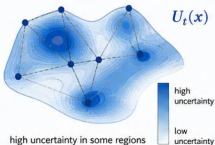


Variance-bias active selection within node-free simplices

Quijote GP posterior variance

current training set D_t

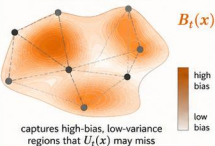
$U_t(x)$



KUN bias proxy

bias measured on matched standard-geometry simplices

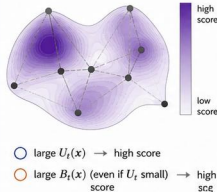
$B_t(x)$



Normalized acquisition score

$$S_t(x) = \left[\left(\frac{U_t(x)}{s_{U,t}} \right)^2 + \left(\frac{\lambda B_t(x)}{s_{B,t}} \right)^2 \right]^{1/2}$$

balances uncertainty and bias



- large $U_t(x)$ → high score
- large $B_t(x)$ (even if U_t small) → high score

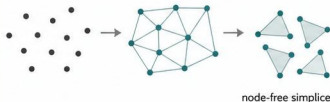
continuous score field over parameter space

Delaunay simplex search (one point per round)

D_t

$\mathcal{T}(D_t)$

$\{\Delta_m\}$



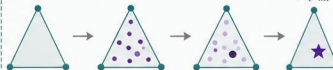
node-free simplices

stage 0

QMC refinement

polish

max over $\text{int}(\Delta_m)$



interior contains no training node

candidate probes inside simplex

evaluate $S_t(x)$ continuously inside each simplex
select best point

Selected new point

★ x_{t+1}

add to training set

$D_{t+1} = D_t \cup \{x_{t+1}\}$



query Quijote truth

retrain emulator
next round

one point
per round

● Variance (U_t):

explores where the emulator is uncertain.

● Bias proxy (B_t):

targets regions with expected modeling bias.

● Score (S_t):

combines both signals for balanced acquisition.

● Simplex search:

continuous optimization inside node-free Delaunay simplices.

★ Outcome:

one high-score point x_{t+1} is selected and added each round.

Selection is not based on uncertainty alone — the bias proxy guides sampling toward hard-to-model regions.